

2 - PID Control

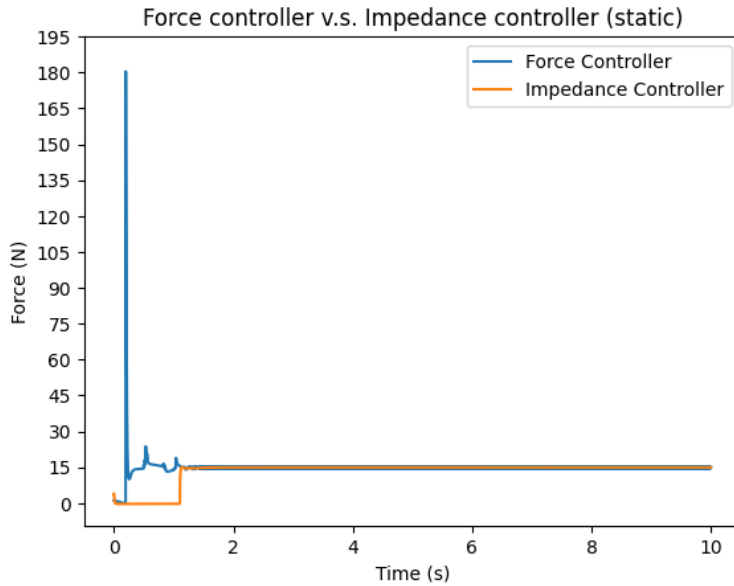


Figure 1: Control with a Fixed Wall

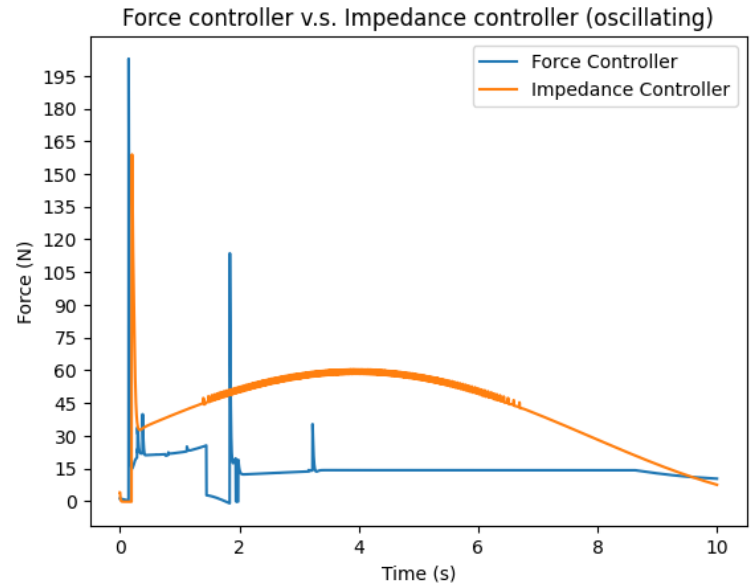


Figure 2: Control with a Moving Wall

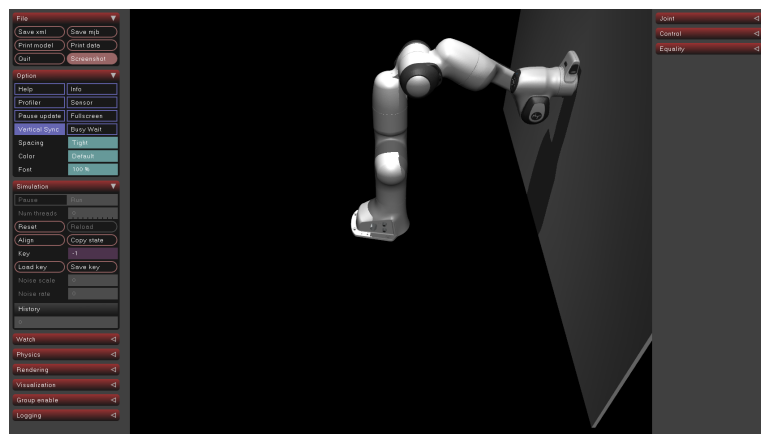


Figure 3: Whiteboard near maximum amplitude

Differences in behavior between the force controller and the impedance controller

The force controller directly commands the desired force, while the impedance controller generates force from position and velocity errors. As shown in Figure 1, the force controller overshoots and oscillates a few times before settling, while the impedance controller is steadier.

Observation and analysis on how the robot behaves under this disturbance

As shown in Figure 2, the force controller has several large transient spikes but then returns near the target force. The impedance controller behaves like a spring-damper system: it produces a higher force when the wall moves closer (such as in Figure 3) and a lower force when the wall moves farther away, and it also has an initial overshoot. This occurs because the impedance controller uses a virtual spring and damper, which allow it to respond compliantly to the disturbance.

3 - Kinematics

End-effector poses for q_1 , q_2 , and q_3 .

- q_1 end-effector pose:

$$\mathbf{t} = \begin{bmatrix} 0.088 \\ 0 \\ 0.926 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

- q_2 end-effector pose:

$$\mathbf{t} = \begin{bmatrix} 0.157 \\ -0.103 \\ 0.936 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 0.649 & 0.759 & 0.052 \\ 0.755 & -0.651 & 0.073 \\ 0.089 & -0.008 & -0.996 \end{bmatrix}$$

- q_3 end-effector pose:

$$\mathbf{t} = \begin{bmatrix} 0.401 \\ 0.087 \\ 0.855 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 0.980 & -0.181 & -0.081 \\ -0.174 & -0.593 & -0.786 \\ 0.095 & 0.785 & -0.612 \end{bmatrix}$$

IK joint angles for (R_g, t_g)

- IK joint angles (rad):

$$\mathbf{q} = [0.180637 \quad 0.257370 \quad -0.209686 \quad -2.082814 \quad -0.032940 \quad 3.907053 \quad -3.064487]$$

Robot at the IK configuration

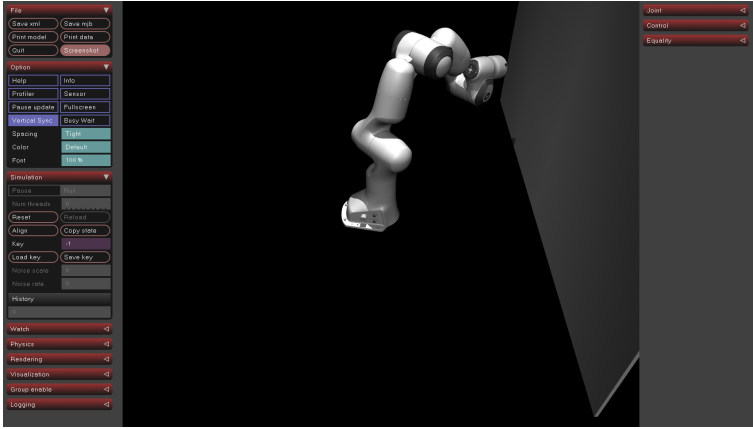


Figure 4: Franka arm with IK joint targets.

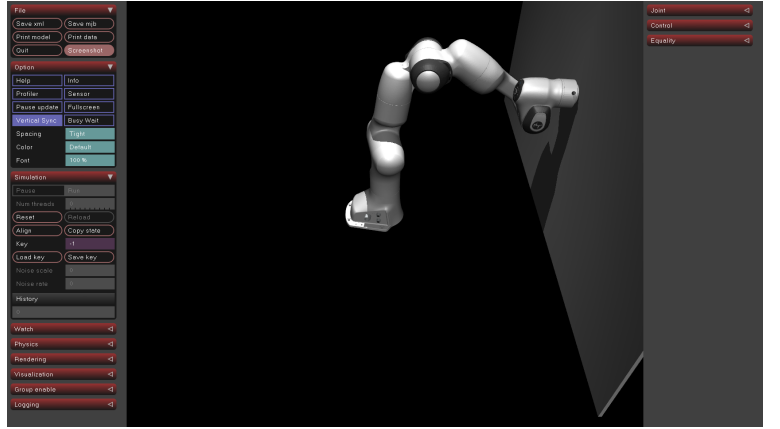


Figure 5: Same IK joint targets with tuned gains.

With the original PD gains ($K_p = 1000$, $K_d = 1000$), some joints did not settle at their IK target angles because the actuator torques saturated (Figure 4). After reducing the gains ($K_p = 100$, $K_d = 20$), the robot moved closer to the desired joint configuration (Figure 5).